Analyzing the Impact of Car Features on Price and Profitability

Description:

In this project, I analyzed the “Car Features and MSRP” dataset to understand how different vehicle features affect car pricing, fuel efficiency, popularity, and overall market demand. The main objective of this project was to identify the key factors that influence a car’s MSRP and help manufacturers make better pricing and product development decisions.

The automotive industry has become highly competitive over the years, with increasing demand for fuel-efficient, technologically advanced, and performance-oriented vehicles. Because of this, manufacturers need to understand which vehicle features are most valuable to customers and which factors contribute most to profitability.

Using Microsoft Excel, I performed complete data cleaning, exploratory data analysis, regression analysis, correlation analysis, and interactive dashboard creation to generate business insights from the dataset.

<u>Business Problem</u> -

The project focuses on answering the following business question:

“How can car manufacturers optimize pricing and product development decisions to maximize profitability while meeting consumer demand?”

To answer this question, I analyzed the relationship between:

- Engine power and MSRP
- Fuel efficiency and engine cylinders
- Vehicle popularity and market category
- Body style and pricing
- Transmission type and MSRP
- Horsepower, MPG, and price across different brands

The project also includes an interactive dashboard that allows users to analyze data dynamically using slicers and filters.

Approach:

<u>Understanding Dataset</u> –

The dataset contains information about different car models and their specifications, including engine performance, fuel type, transmission type, vehicle style, fuel efficiency, popularity, and MSRP. By analyzing these variables, I was able to study how different car features influence pricing, fuel efficiency, and consumer demand.

Data Cleaning Process -

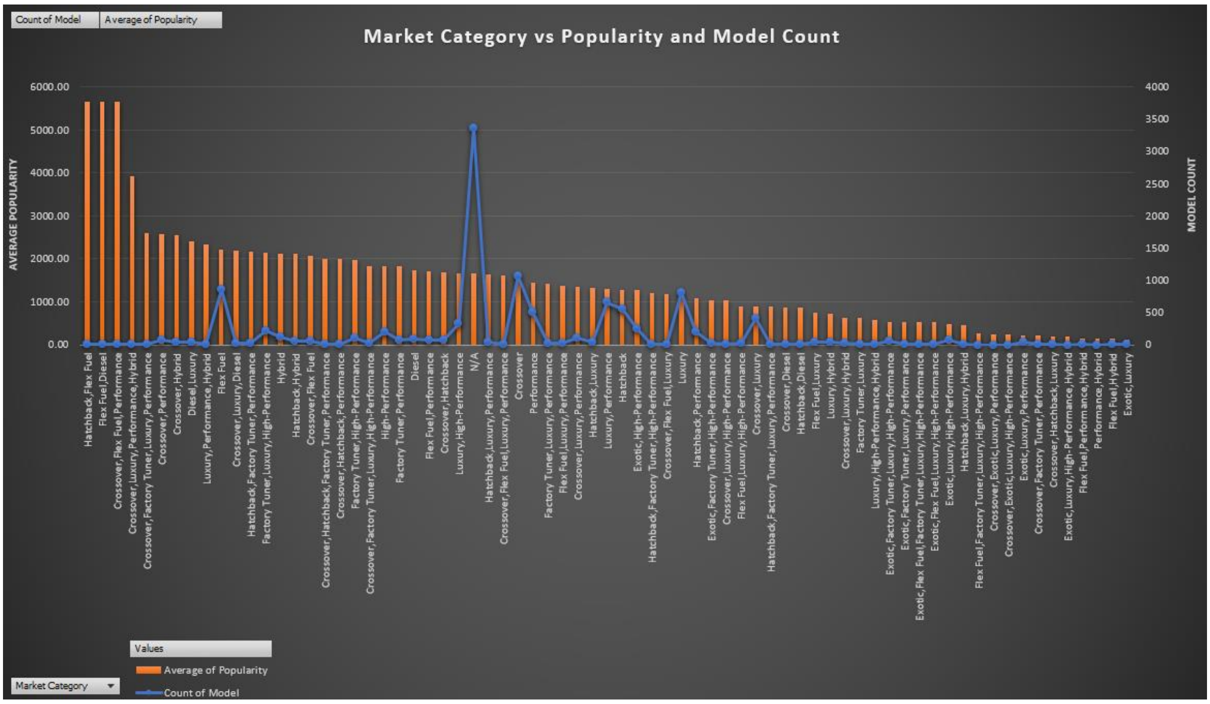
Before starting the analysis, I cleaned the dataset to improve accuracy and consistency. The following steps were performed:

- Removed blank cells and missing values
- Removed duplicate records
- Corrected formatting issues
- Replaced unknown values using mode values
- Formatted numerical columns properly
- Organized the dataset for analysis and dashboard creation

Analysis –

Task 1: Popularity Across Market Categories

Row Labels	Count of Model	Average of Popularity
Hatchback,Flex Fuel	7	5657.00
Flex Fuel,Diesel	16	5657.00
Crossover,Flex Fuel,Performance	6	5657.00
Crossover,Luxury,Performance,Hybrid	2	3916.00
Crossover,Factory Tuner,Luxury,Performance	5	2607.40
Crossover,Performance	69	2585.96
Crossover,Hybrid	42	2563.38

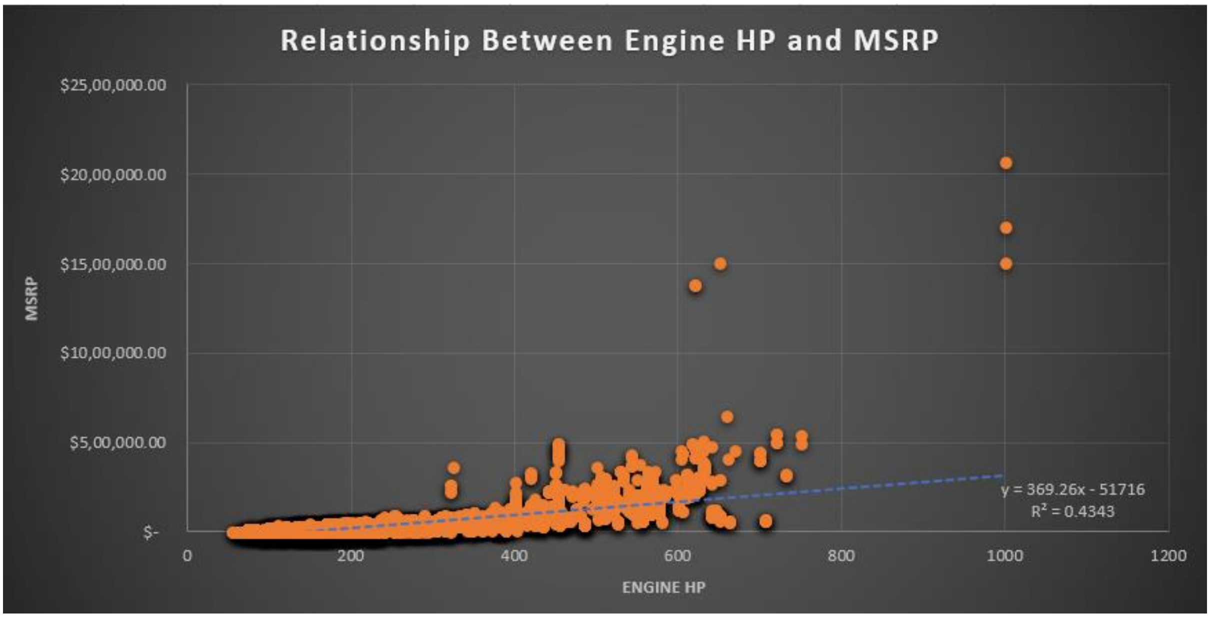


The analysis showed that categories such as “Hatchback, Flex Fuel,” “Flex Fuel, Diesel,” and “Crossover, Performance” had some of the highest average popularity scores in the dataset.

The “Flex Fuel” category alone contained a very high number of car models, indicating strong manufacturer presence and consumer interest in flexible fuel technology. Performance and luxury crossover categories also demonstrated high popularity, suggesting that consumers prefer vehicles that combine performance, comfort, and practicality.

Overall, the analysis indicates that fuel-efficient and performance-oriented vehicle categories attract higher customer attention in the automotive market.

Task 2: Relationship Between Engine Power and MSRP



The scatter plot showed a clear positive relationship between engine horsepower and MSRP. Vehicles with higher engine horsepower generally had significantly higher prices.

Cars with lower horsepower values were concentrated in the lower price range, while high-performance vehicles with higher horsepower occupied the upper range of the pricing spectrum.

This indicates that engine performance is one of the major factors affecting vehicle pricing, especially in luxury and sports car segments.

Task 3: Important Features Affecting MSRP

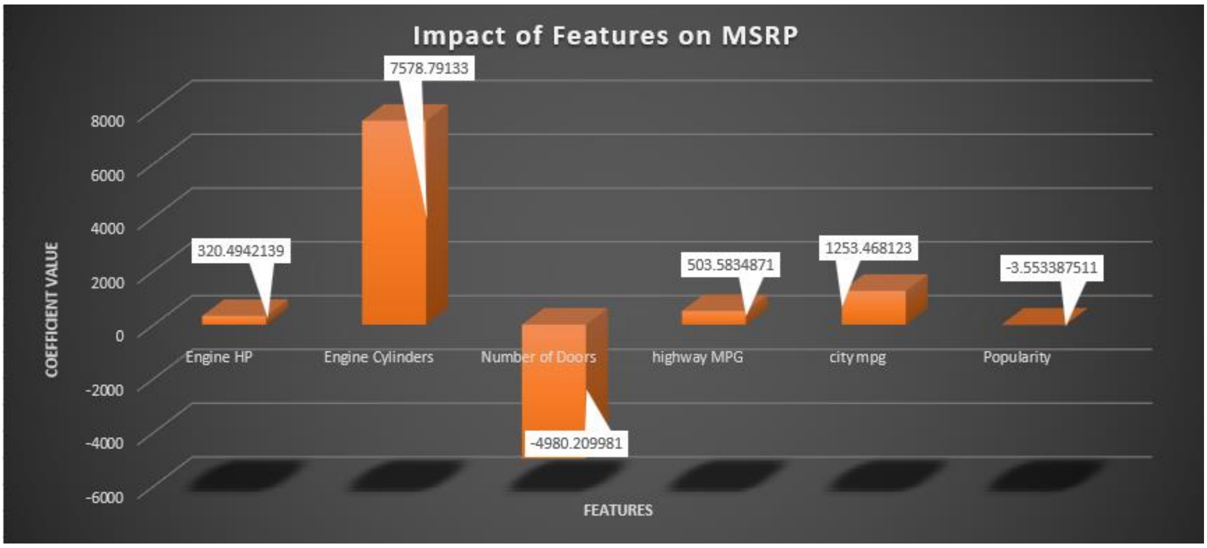
SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.683387437
R Square	0.46701839
Adjusted R Square	0.466730032
Standard Error	45078.99614
Observations	11097

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	6	1.9747E+13	3.29117E+12	1619.578688	0
Residual	11090	2.25362E+13	2032115893		
Total	11096	4.22832E+13			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	-97167.87274	3898.078612	-24.92711985	1.7136E-133	-104808.8004	-89526.94512	-104808.8004	-89526.94512
Engine HP	320.4942139	6.3774589	50.25421863	0	307.9932598	332.995168	307.9932598	332.995168
Engine Cylinders	7578.79133	461.2602827	16.43061762	5.88653E-60	6674.639109	8482.94355	6674.639109	8482.94355
Number of Doors	-4980.209981	496.4047724	-10.03255863	1.38198E-23	-5953.251655	-4007.168308	-5953.251655	-4007.168308
highway MPG	503.5834871	109.2773107	4.608307836	4.10488E-06	289.3805157	717.7864585	289.3805157	717.7864585
city mpg	1253.468123	125.6629389	9.974843293	2.46287E-23	1007.146405	1499.789841	1007.146405	1499.789841
Popularity	-3.553387511	0.297352947	-11.95006655	1.02989E-32	-4.136252193	-2.97052283	-4.136252193	-2.97052283



The regression model explained approximately 46.7% of the variation in vehicle MSRP, indicating a moderate relationship between the selected variables and pricing.

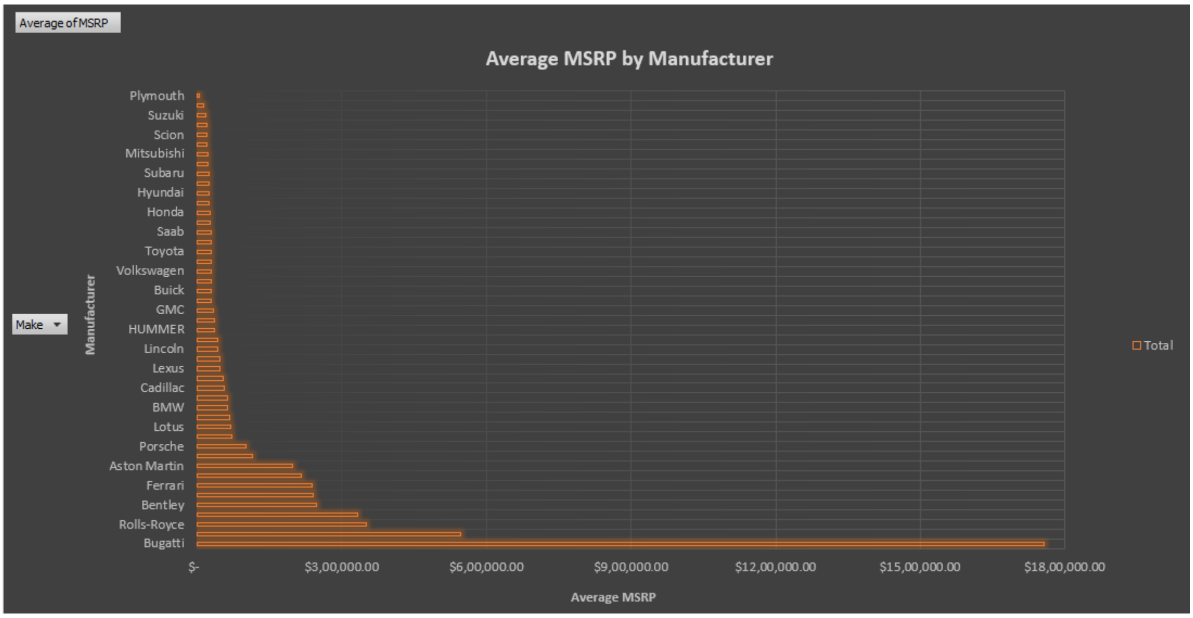
The coefficient analysis showed:

- Engine HP had the strongest positive impact on MSRP.
- Engine Cylinders also positively affected pricing.
- City MPG and Highway MPG showed positive relationships with MSRP.
- Number of Doors showed a negative relationship with price.
- Popularity also showed a slight negative impact on MSRP.

The results indicate that performance-related features such as horsepower and engine size contribute significantly to higher vehicle prices.

<u>Task 4: Average Price Across Manufacturers</u>

Row Labels	Average of MSRP
Bugatti	$ 17,57,223.67
Maybach	$ 5,46,221.88
Rolls-Royce	$ 3,51,130.65
Lamborghini	$ 3,31,567.31
Bentley	$ 2,47,169.32
McLaren	$ 2,39,805.00
Ferrari	$ 2,37,383.82
Spyker	$ 2,14,990.00
Aston Martin	$ 1,98,123.46
Maserati	$ 1,13,684.49
Porsche	$ 1,01,622.40
Mercedes-Benz	$ 72,135.03



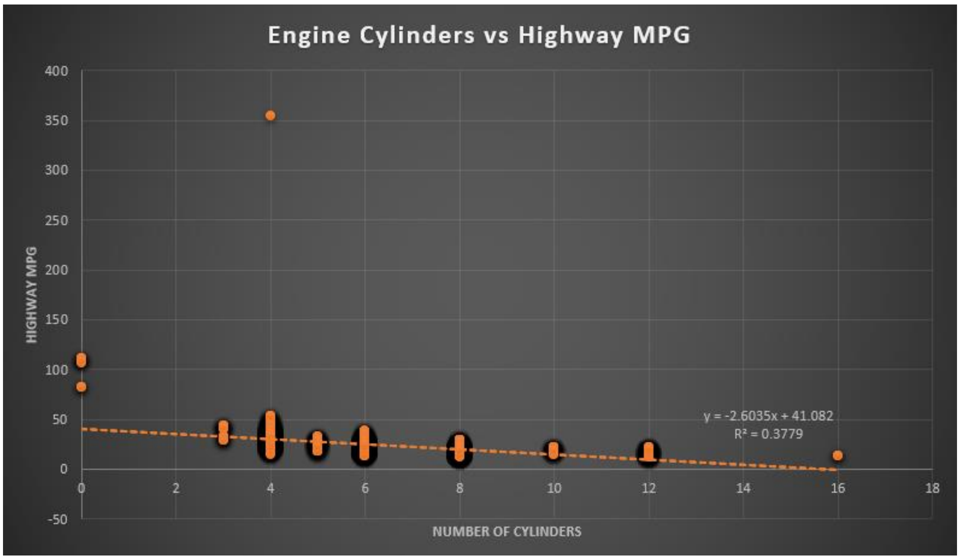
The analysis revealed that Bugatti had the highest average MSRP in the dataset, followed by Maybach, Rolls-Royce, Lamborghini, and Bentley.

Luxury and exotic car manufacturers consistently maintained extremely high average vehicle prices compared to mainstream brands. Premium manufacturers such as Ferrari, Aston Martin, Maserati, and Porsche also showed significantly higher average MSRPs.

This indicates that brand value, luxury positioning, exclusivity, and performance strongly influence vehicle pricing in the automotive industry.

Task 5: Fuel Efficiency vs Engine Cylinders

CORRELATION
-0.614703148



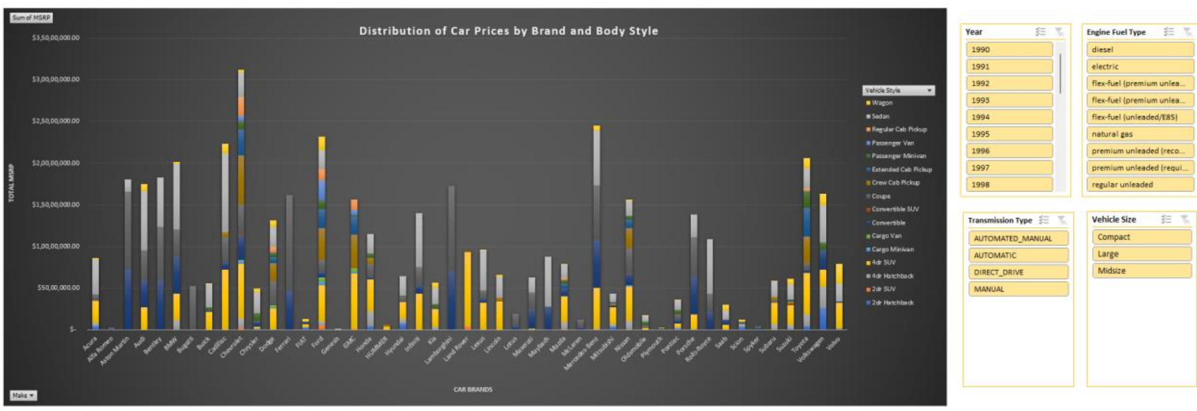
The correlation analysis showed a moderately strong negative relationship between engine cylinders and highway MPG.

As the number of engine cylinders increased, highway fuel efficiency generally decreased. Vehicles with larger engines consumed more fuel and delivered lower MPG values compared to vehicles with smaller engines.

This suggests that manufacturers aiming to improve fuel efficiency should focus on smaller and more efficient engine designs.

Dashboard Analysis –

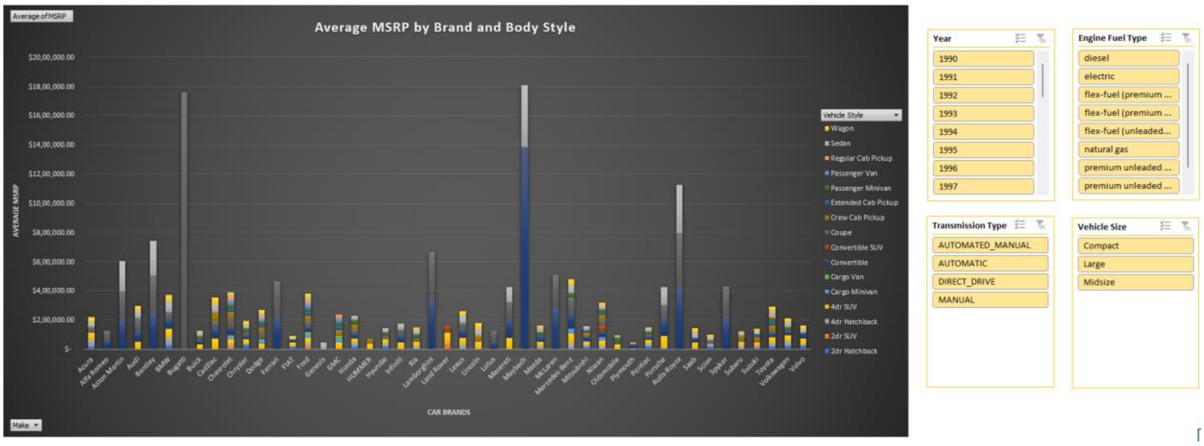
Dashboard Task 1: Distribution of Car Prices by Brand and Body Style



The chart showed that luxury brands contributed significantly to total MSRP distribution across multiple body styles. SUVs and coupe categories generated higher overall pricing compared to smaller vehicle styles.

The dashboard also showed that premium manufacturers dominate higher-priced vehicle segments.

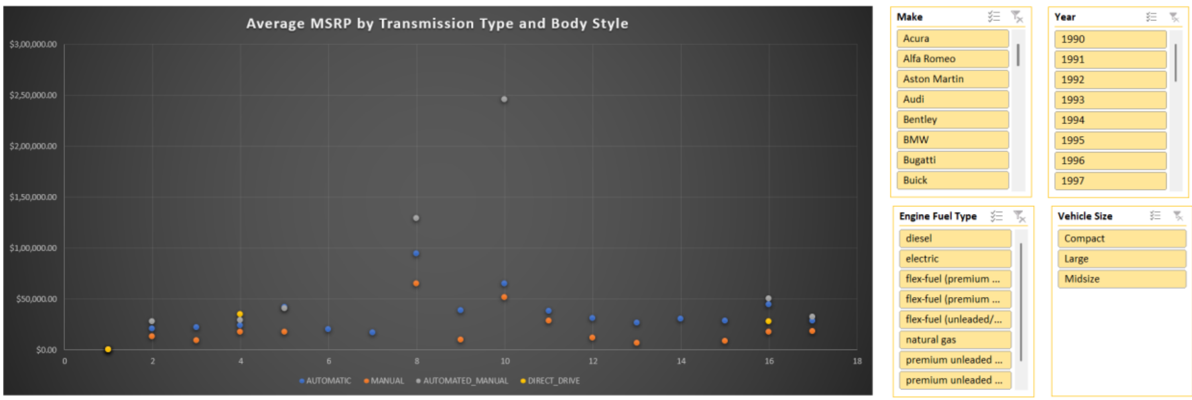
Dashboard Task 2: Average MSRP by Brand and Body Style



The chart revealed that luxury brands consistently maintained higher average MSRPs across most body styles. Coupe and SUV body styles generally had higher average prices compared to hatchbacks and sedans.

This indicates strong customer demand for premium SUVs and performance-oriented vehicles.

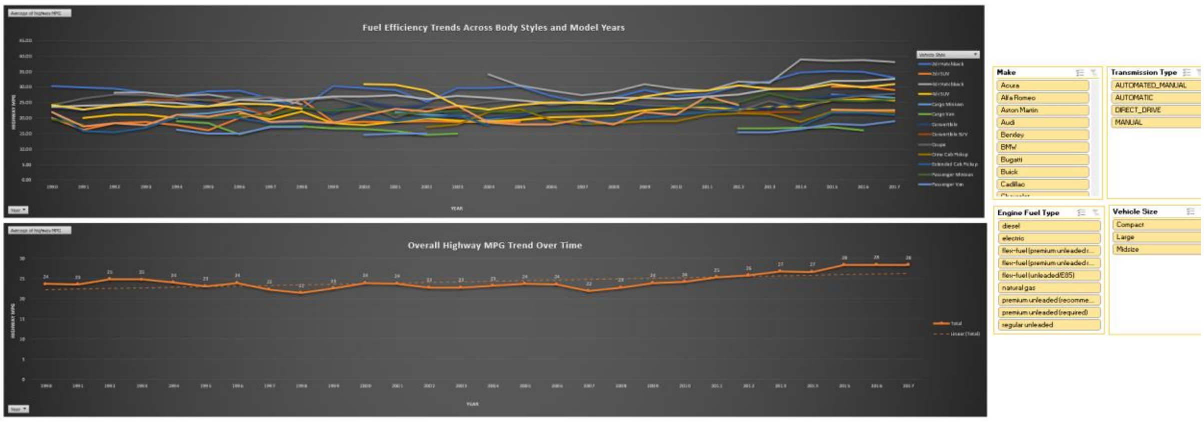
Dashboard Task 3: Impact of Transmission Type on MSRP



The scatter plot analysis showed that automatic and automated manual transmission vehicles generally had higher average MSRPs compared to manual transmission vehicles.

Advanced transmission systems were more commonly associated with premium and high-performance vehicles.

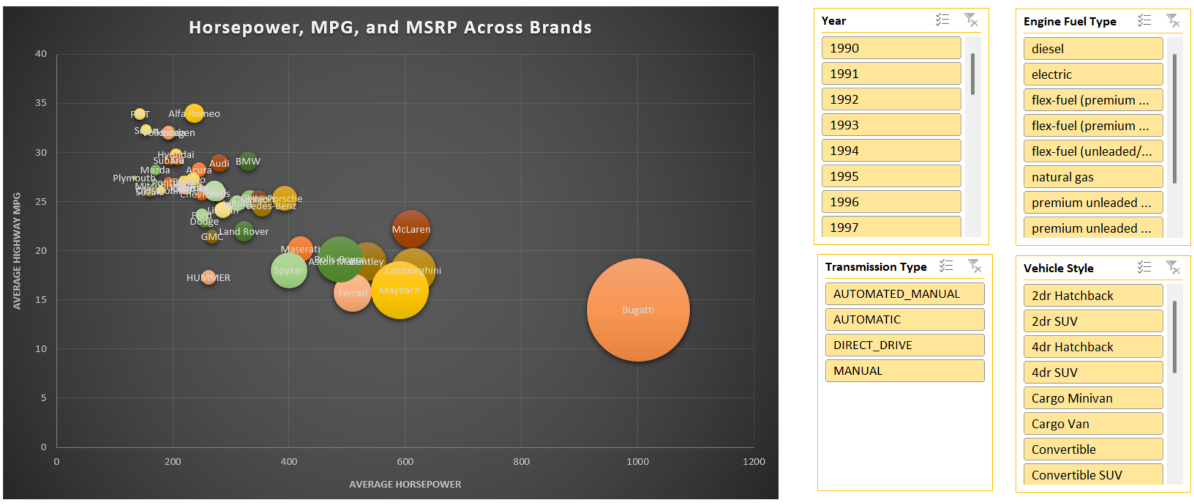
Dashboard Task 4: Fuel Efficiency Trends Across Body Styles and Years



The line chart showed that fuel efficiency improved gradually across most vehicle body styles over the years.

Sedans consistently demonstrated better highway MPG compared to SUVs and larger vehicle categories. The trend reflects increasing industry focus on fuel-efficient technologies and improved engine performance.

Dashboard Task 5: Horsepower, MPG, and MSRP Across Brands



The bubble chart showed that luxury and performance brands generally had:

- Higher horsepower
- Larger bubble sizes (higher MSRP)
- Lower fuel efficiency

Economy-focused brands demonstrated:

- Better MPG
- Lower pricing
- Moderate horsepower

This highlights the trade-off between vehicle performance, fuel efficiency, and pricing across different automotive brands.

<u>Final Dashboard</u> –

Tech Stack Used:

Microsoft Excel Worksheet (2019) – Used for data cleaning, analysis, and dashboard creation.

Excel Pivot Tables and Pivot Charts – Used for data summarization and visualization.

Excel Charts – Used for scatter plots, line charts, bar charts, and bubble charts.

Excel Statistical Functions and Data Analysis ToolPak – Used for correlation and regression analysis.

Excel Slicers and Filters – Used for creating an interactive dashboard.

Microsoft Word / PDF – Used for project documentation and report preparation.

Insight:

From this project, I found that engine performance and brand positioning play a major role in determining car prices. Vehicles with higher horsepower and larger engines generally had higher MSRPs, especially luxury and performance-oriented brands such as Bugatti, Lamborghini, Bentley, and Rolls-Royce.

I also observed that fuel efficiency decreases as the number of engine cylinders increases. Smaller-engine vehicles showed better MPG values, while high-performance vehicles focused more on power than fuel economy.

The analysis further showed that luxury SUVs and coupe-style vehicles contribute significantly to higher pricing segments. Automatic and advanced transmission systems were also commonly associated with higher-priced vehicles.

Result:

This project helped me understand how data analytics can be used to solve real-world business problems in the automotive industry. Through regression analysis, correlation analysis, Pivot Tables, and dashboard visualization, I identified important relationships between vehicle features, pricing, performance, and fuel efficiency.

The analysis showed that engine horsepower and engine cylinders significantly influence vehicle pricing, while fuel efficiency decreases as the number of engine cylinders increases. Luxury and performance brands generally maintain higher MSRPs, while economy-focused brands prioritize affordability and fuel efficiency.

The interactive dashboard created in Excel allows users to explore the data dynamically using slicers and filters, making the analysis more effective and visually understandable.

Overall, this project improved my understanding of data analysis, visualization, and dashboard development in Excel, while also showing how data-driven insights can support pricing strategies, product development, and decision-making in the automotive industry.

Drive link:

https://drive.google.com/drive/folders/1dYKIt7Y8SxBLgh0_QgSq-kNTZZk2fMHK?usp=sharing

Click here for Excel Sheet file Car Dataset

Click here Video Presentation